Audio Visualiser DSP Board

https://github.com/drewsum/audio_visualiser

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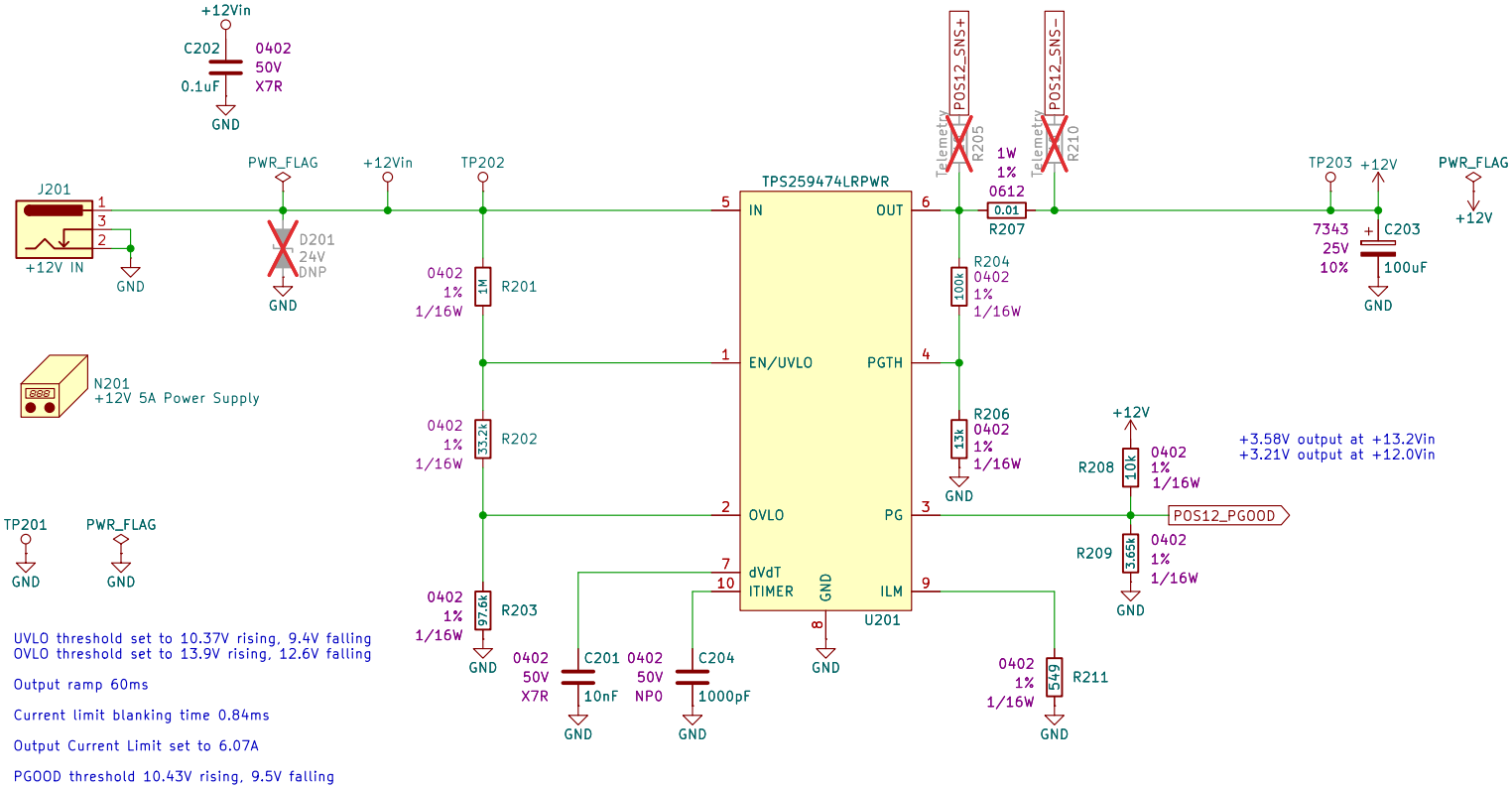
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Sheet: /		
File: Audio_Visualiser_DSP.kicad_sch		
Title:		
Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 1/21

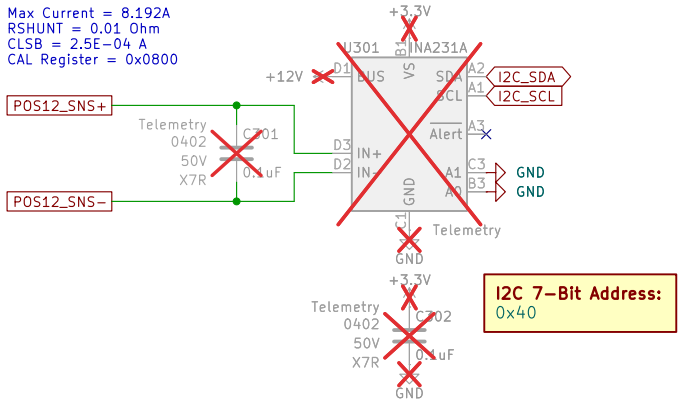


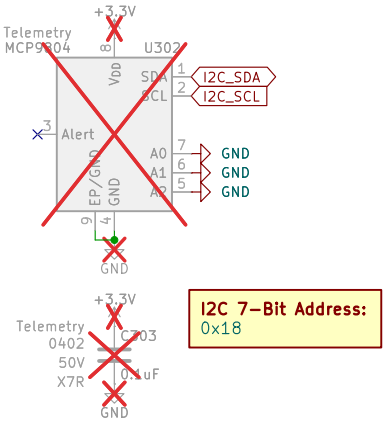
Load	Draw	Quantity	Total
LED Panel VLED 256 * 0.02A 8 3V = 15.36W / 12V = 1.28A / 80% efficiency	1.6A	2	3.2A
LED Panel POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	2	0.688A
Local POS3P3 1A * 3.3V / 12V / 80% efficiency	0.344A	1	0.344A
Total		50.784W	4.232A

Sheet: /12V Power Input/
File: POS12_power_input.kicad_sch

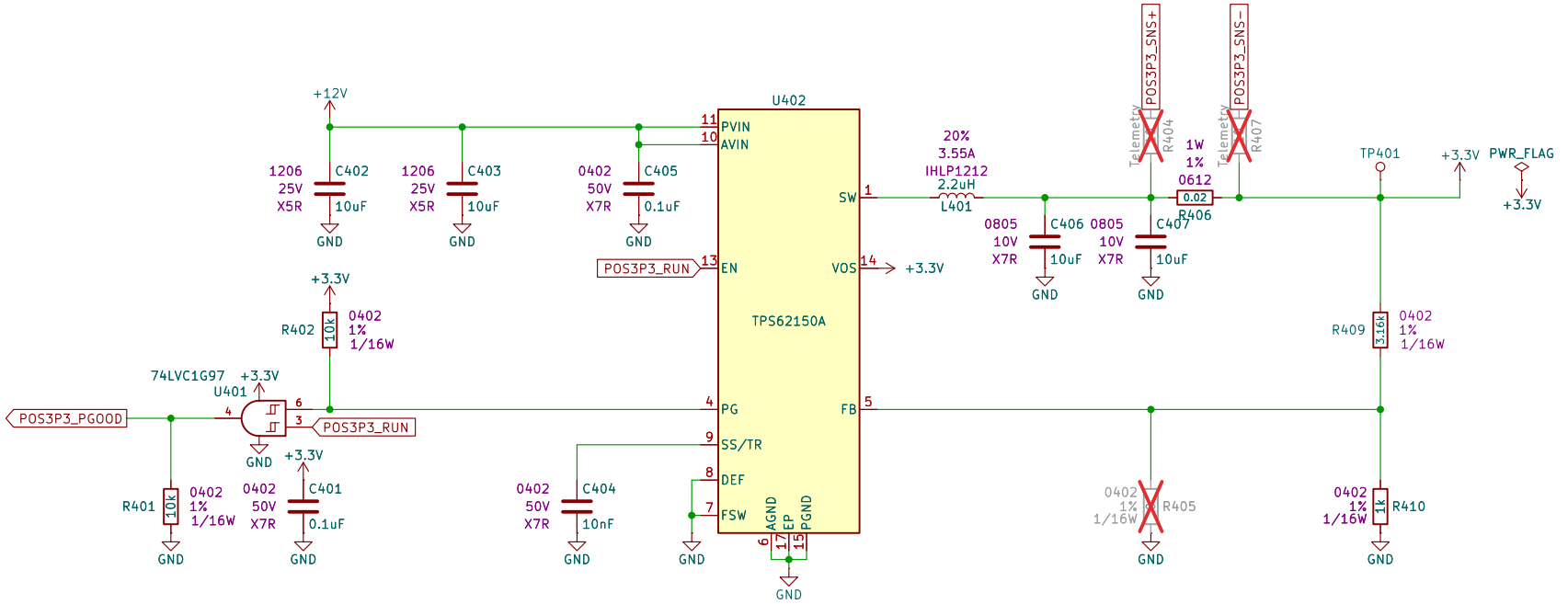
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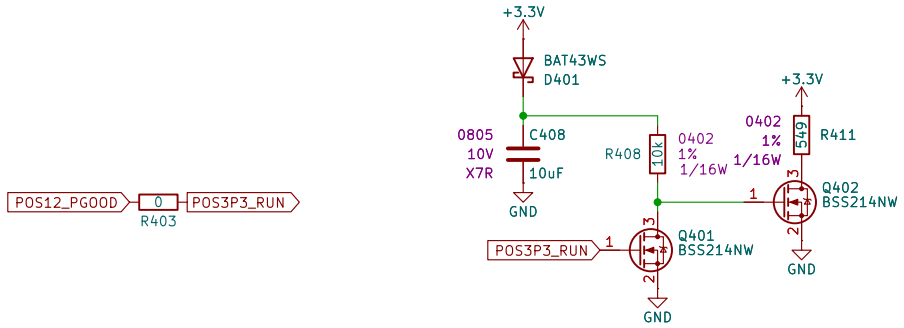
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Sheet: /12V Telemetry/		
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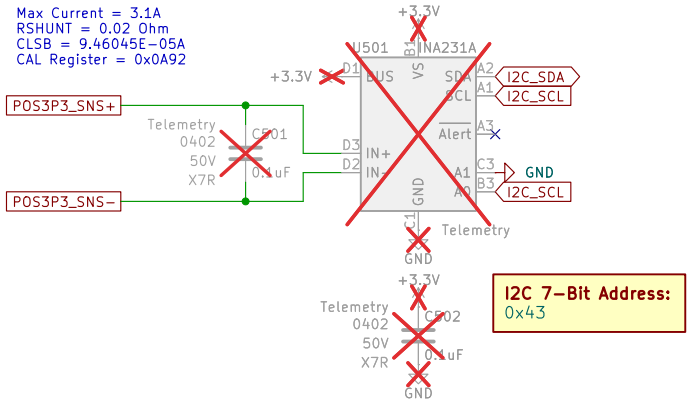


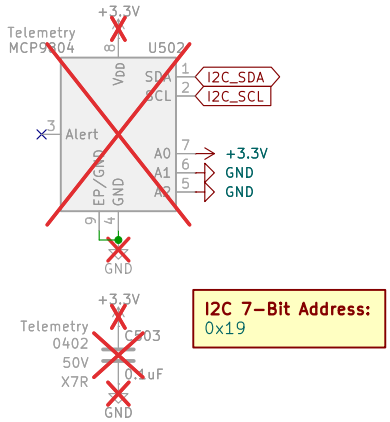
Load	Draw	Quantity	Total
dsPIC	50mA	1	50mA
8MHz dsPIC Clock	7mA	1	7mA
MCP6N11	1.1mA	6	6.6mA
MCP6024	5.4mA	3	16.2mA
Analog Switch	0.05mA	1	0.05mA
Digipot	0.6mA	1	0.6mA
1.6V Reference	0.2mA	1	0.2mA
PCA9536 GPIO Expander	0.29mA	1	0.29mA
I2C Buffers	8.3mA	3	24.9mA
Octal Buffers	1mA	2	2mA
LVC1G19s	0.01mA	21	0.21mA
Rotary Encoder Module	50mA	1	50mA
LEDs	1.3mA	11	14.3mA
INA231	0.42mA	2	0.84mA
MCP9804	0.4mA	3	1.2mA
ETC	0.36mA	1	0.36mA
Cap Touch Sensor	0.729mA	1	0.729mA
Total		0.579W	175.48mA

Sheet: /3.3V Power Supply/
File: POS3P3_power_supply.kicad_sch

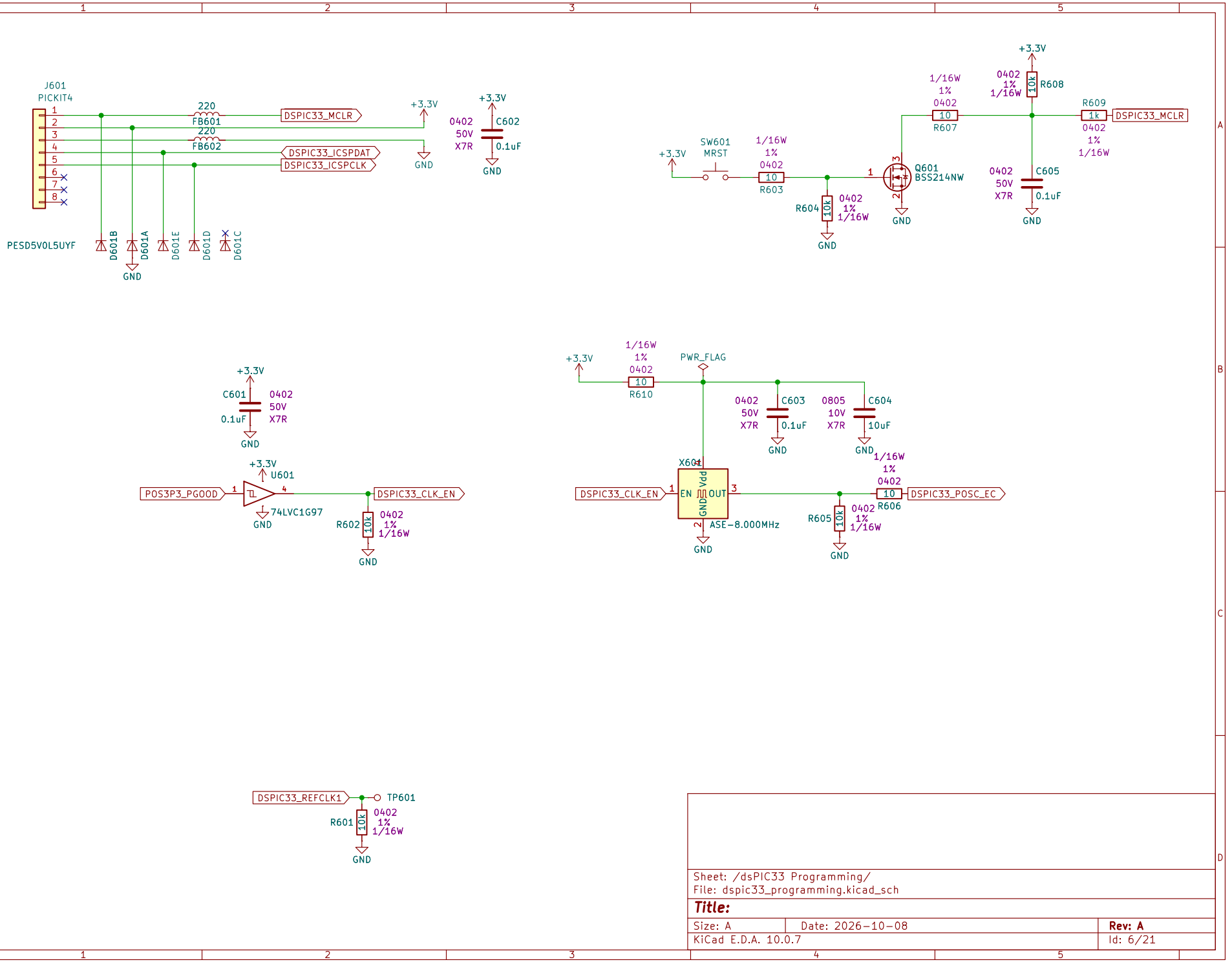
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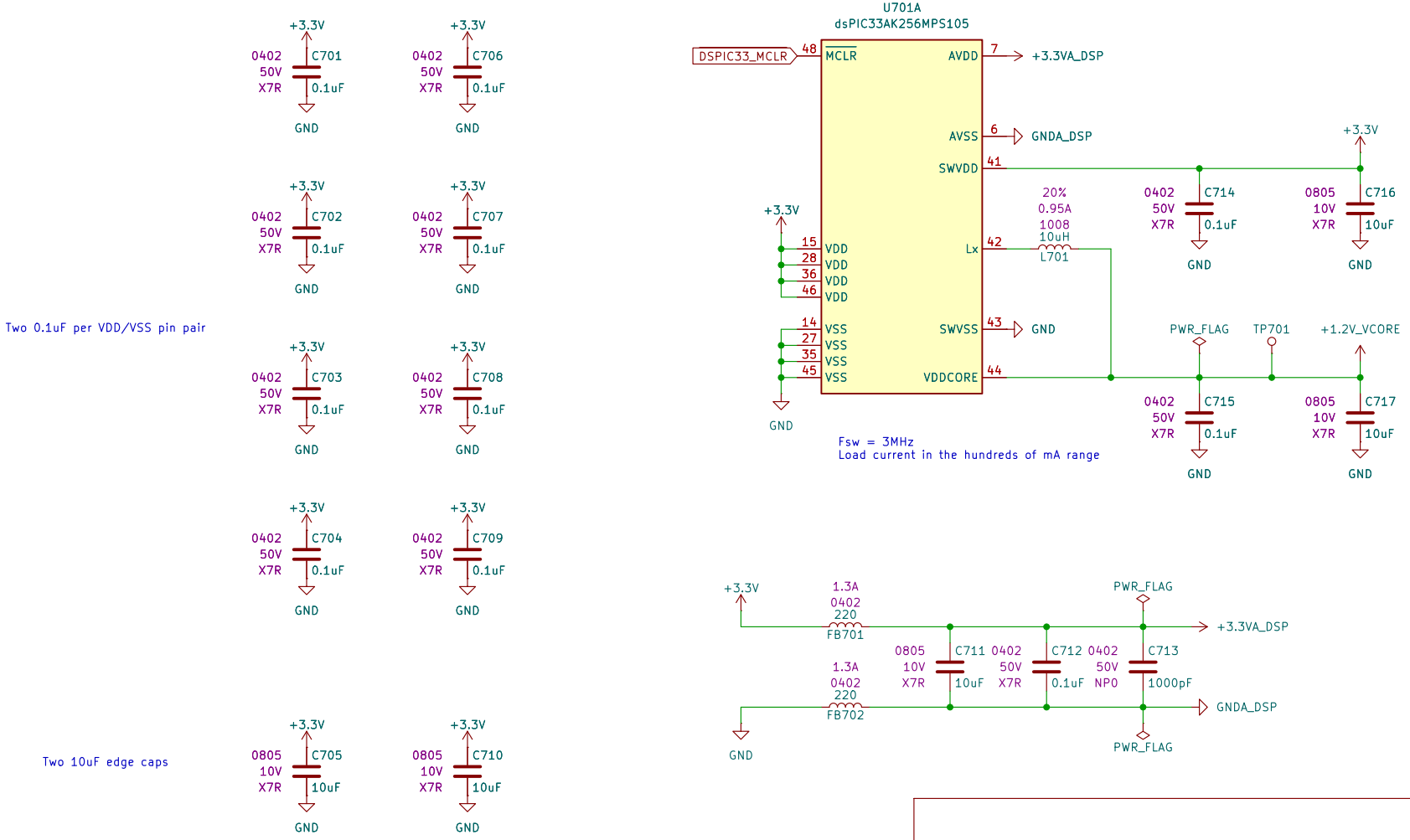
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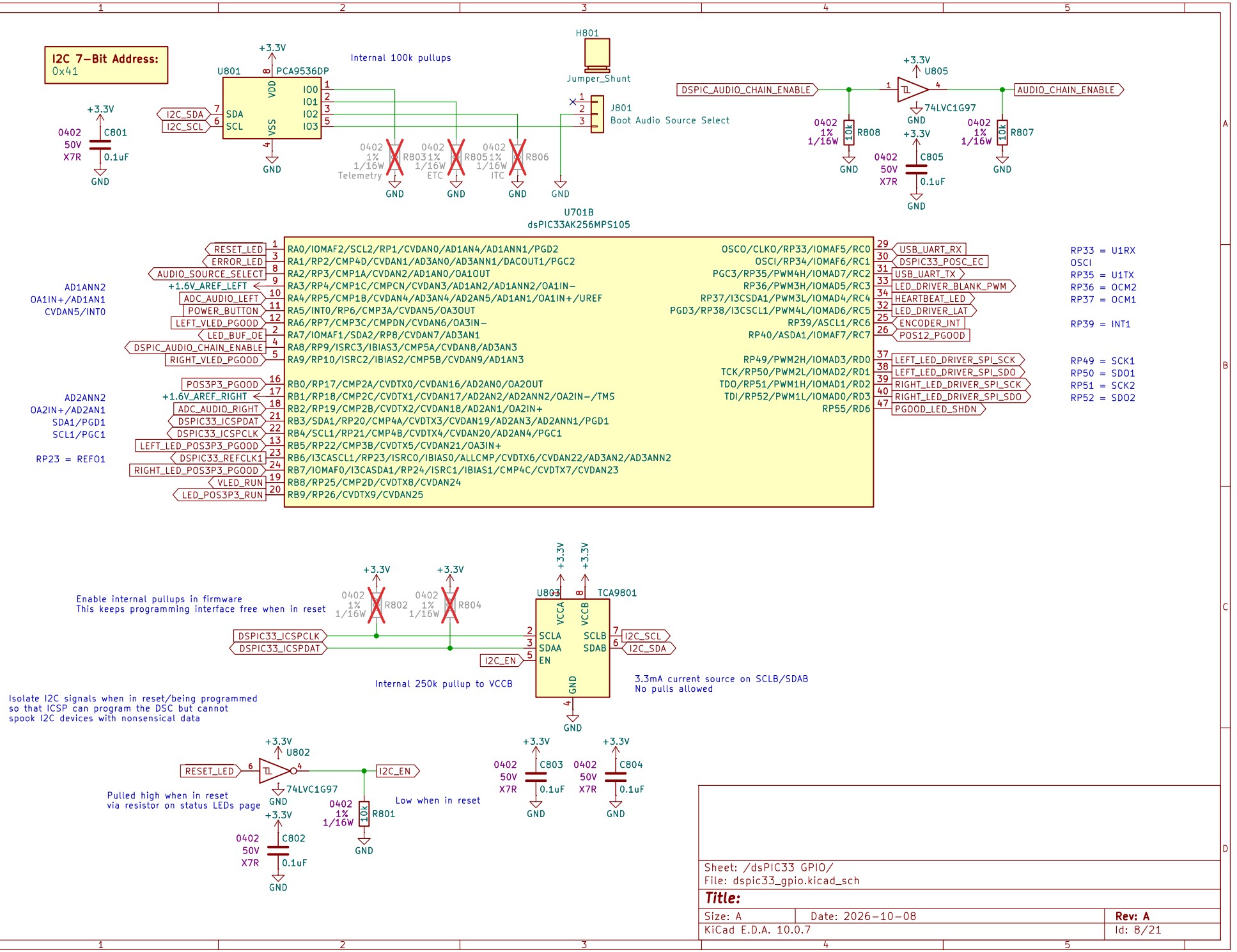


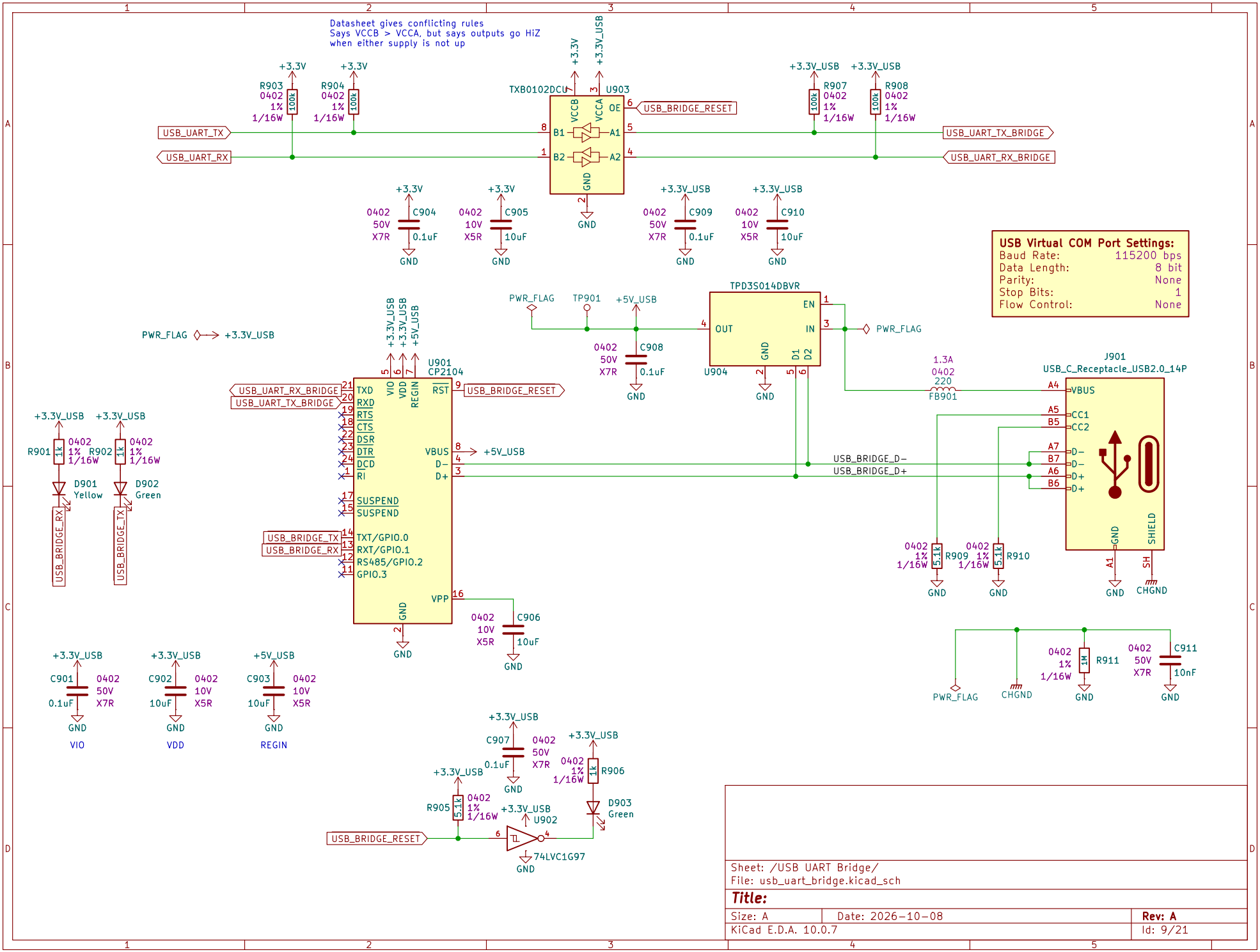
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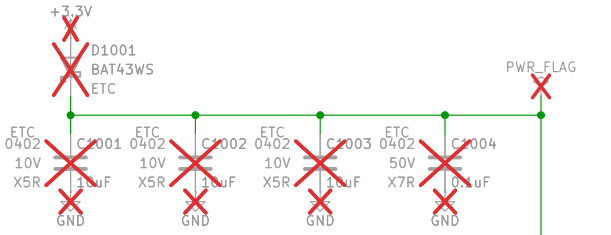

J601
PICKIT4
1
2
3
4
5
6
7
8
220
FB601
220
FB602
DSPIC33_MCLR
DSPIC33_ICSPDAT
DSPIC33_ICSPCLK
+3.3V
GND
+3.3V
0402
50V
X7R
C602
0.1uF
GND
PESD5V0L5UYF
D601B
D601A
D601E
D601D
D601C
GND
+3.3V
1/16W
1%
0402
10
R607
0402
1%
1/16W
10k
R608
R609
1k
DSPIC33_MCLR
0402
1%
1/16W
+3.3V
SW601
MRST
1/16W
1%
0402
10
R603
Q601
BSS214NW
R604
10k
0402
1%
1/16W
GND
0402
50V
X7R
C605
0.1uF
GND
+3.3V
C601
0402
50V
X7R
0.1uF
GND
+3.3V
U601
POS3P3_PGOOD
1
TL
4
DSPIC33_CLK_EN
74LVC1G97
GND
R602
10k
0402
1%
1/16W
GND
+3.3V
1/16W
1%
0402
10
R610
PWR_FLAG
0402
50V
X7R
C603
0.1uF
GND
0805
10V
X7R
C604
10uF
GND
X601
DSPIC33_CLK_EN
1
EN
Vdd
OUT
3
GND
2
ASE-8.000MHz
GND
1/16W
1%
0402
10
R606
DSPIC33_POSC_EC
R605
10k
0402
1%
1/16W
GND
DSPIC33_REFCLK1
TP601
R601
10k
0402
1%
1/16W
GND
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Date: 2026-10-08
Rev: A
KiCad E.D.A. 10.0.7
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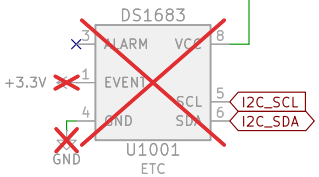


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KiCad E.D.A. 10.0.7		Id: 7/21

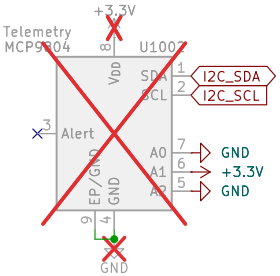

I2C 7-Bit Address:
0x41
U801
PCA9536DP
Internal 100k pullups
H801
Jumper_Shunt
J801
Boot Audio Source Select
R808
R807
R805
R803
R804
R802
R801
C801
C802
C803
C804
C805
74LVC1G97
U805
U802
U803
TCA9801
U701B
dsPIC33AK256MPS105
RA0/IOMAF2/SCL2/RP1/CVDAN0/AD1AN4/AD1ANN1/PGD2
RA1/RP2/CMP4D/CVDAN1/AD3AN0/AD3ANN1/DACOUT1/PGC2
RA2/RP3/CMP1A/CVDAN2/AD1AN0/OA1OUT
RA3/RP4/CMP1C/CMPCN/CVDAN3/AD1AN2/AD1ANN2/OA1IN−
RA4/RP5/CMP1B/CVDAN4/AD3AN4/AD2AN5/AD1AN1/OA1IN+/UREF
RA5/INT0/RP6/CMP3A/CVDAN5/OA3OUT
RA6/RP7/CMP3C/CMPDN/CVDAN6/OA3IN−
RA7/IOMAF1/SDA2/RP8/CVDAN7/AD3AN1
RA8/RP9/ISRC3/IBIAS3/CMP5A/CVDAN8/AD3AN3
RA9/RP10/ISRC2/IBIAS2/CMP5B/CVDAN9/AD1AN3
RB0/RP17/CMP2A/CVDTX0/CVDAN16/AD2AN0/OA2OUT
RB1/RP18/CMP2C/CVDTX1/CVDAN17/AD2AN2/AD2ANN2/OA2IN−/TMS
RB2/RP19/CMP2B/CVDTX2/CVDAN18/AD2AN1/OA2IN+
RB3/SDA1/RP20/CMP4A/CVDTX3/CVDAN19/AD2AN3/AD2ANN1/PGD1
RB4/SCL1/RP21/CMP4B/CVDTX4/CVDAN20/AD2AN4/PGC1
RB5/RP22/CMP3B/CVDTX5/CVDAN21/OA3IN+
RB6/I3CASCL1/RP23/ISRC0/IBIAS0/ALLCMP/CVDTX6/CVDAN22/AD3AN2/AD3ANN2
RB7/IOMAF0/I3CASDA1/RP24/ISRC1/IBIAS1/CMP4C/CVDTX7/CVDAN23
RB8/RP25/CMP2D/CVDTX8/CVDAN24
RB9/RP26/CVDTX9/CVDAN25
OSCO/CLKO/RP33/IOMAF5/RC0
OSCI/RP34/IOMAF6/RC1
PGC3/RP35/PWM4H/IOMAD7/RC2
RP36/PWM3H/IOMAD5/RC3
RP37/I3CSDA1/PWM3L/IOMAD4/RC4
PGD3/RP38/I3CSCL1/PWM4L/IOMAD6/RC5
RP39/ASCL1/RC6
RP40/ASDA1/IOMAF7/RC7
RP49/PWM2H/IOMAD3/RD0
TCK/RP50/PWM2L/IOMAD2/RD1
TDO/RP51/PWM1H/IOMAD1/RD2
TDI/RP52/PWM1L/IOMAD0/RD3
RP55/RD6
RP33 = U1RX
OSCI
RP35 = U1TX
RP36 = OCM2
RP37 = OCM1
RP39 = INT1
RP49 = SCK1
RP50 = SDO1
RP51 = SCK2
RP52 = SDO2
RP23 = REFO1
AD1ANN2
OA1IN+/AD1AN1
CVDAN5/INT0
AD2ANN2
OA2IN+/AD2AN1
SDA1/PGD1
SCL1/PGC1
Enable internal pullups in firmware
This keeps programming interface free when in reset
Internal 250k pullup to VCCB
3.3mA current source on SCLB/SDAB
No pulls allowed
Isolate I2C signals when in reset/being programmed
so that ICSP can program the DSC but cannot
spook I2C devices with nonsensical data
Pulled high when in reset
via resistor on status LEDs page
Low when in reset
Sheet: /dsPIC33 GPIO/
File: dspic33_gpio.kicad_sch
Title:
Size: A
Date: 2026-10-08
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Datasheet gives conflicting rules
Says VCCB > VCCA, but says outputs go HiZ
when either supply is not up
TXB0102DCUR
U903
USB_UART_TX
USB_UART_RX
USB_UART_TX_BRIDGE
USB_UART_RX_BRIDGE
USB_BRIDGE_RESET
R903 0402 1% 1/16W 100k
R904 0402 1% 1/16W 100k
R907 0402 1% 1/16W 100k
R908 0402 1% 1/16W 100k
C904 0402 50V X7R 0.1uF
C905 0402 10V X5R 10uF
C909 0402 50V X7R 0.1uF
C910 0402 10V X5R 10uF
USB Virtual COM Port Settings:
Baud Rate: 115200 bps
Data Length: 8 bit
Parity: None
Stop Bits: 1
Flow Control: None
TPD3S014DBVR
U904
PWR_FLAG
TP901
+5V_USB
C908 0402 50V X7R 0.1uF
1.3A 0402 220 FB901
J901
USB_C_Receptacle_USB2.0_14P
U901
CP2104
USB_UART_RX_BRIDGE
USB_UART_TX_BRIDGE
USB_BRIDGE_D−
USB_BRIDGE_D+
R901 1k 0402 1% 1/16W
R902 1k 0402 1% 1/16W
D901 Yellow
D902 Green
USB_BRIDGE_RX
USB_BRIDGE_TX
R909 5.1k 0402 1% 1/16W
R910 5.1k 0402 1% 1/16W
C901 0.1uF 0402 50V X7R
C902 10uF 0402 10V X5R
C903 10uF 0402 10V X5R
C906 0402 10V X5R 10uF
VIO
VDD
REGIN
C907 0.1uF 0402 50V X7R
R905 5.1k 0402 1% 1/16W
R906 1k 0402 1% 1/16W
D903 Green
U902
74LVC1G97
R911 1M 0402 1% 1/16W
C911 10nF 0402 50V X7R
CHGND
Sheet: /USB UART Bridge/
File: usb_uart_bridge.kicad_sch
Title:
Size: A
Date: 2026-10-08
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I2C 7-Bit Address:
0x6B



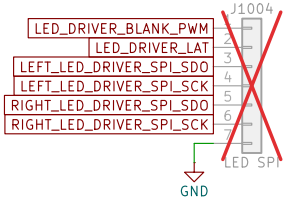


I2C 7-Bit Address:
0x1A

Ambient, place away from power components

Test Points

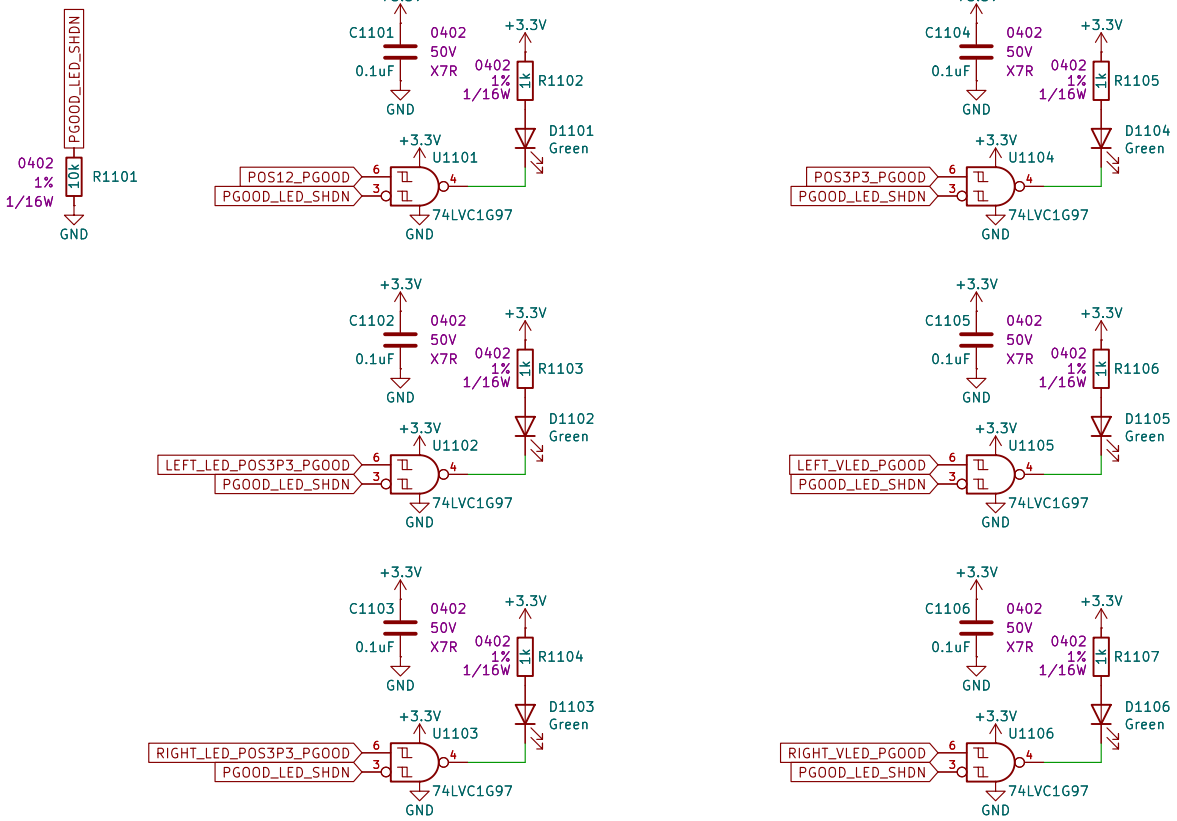




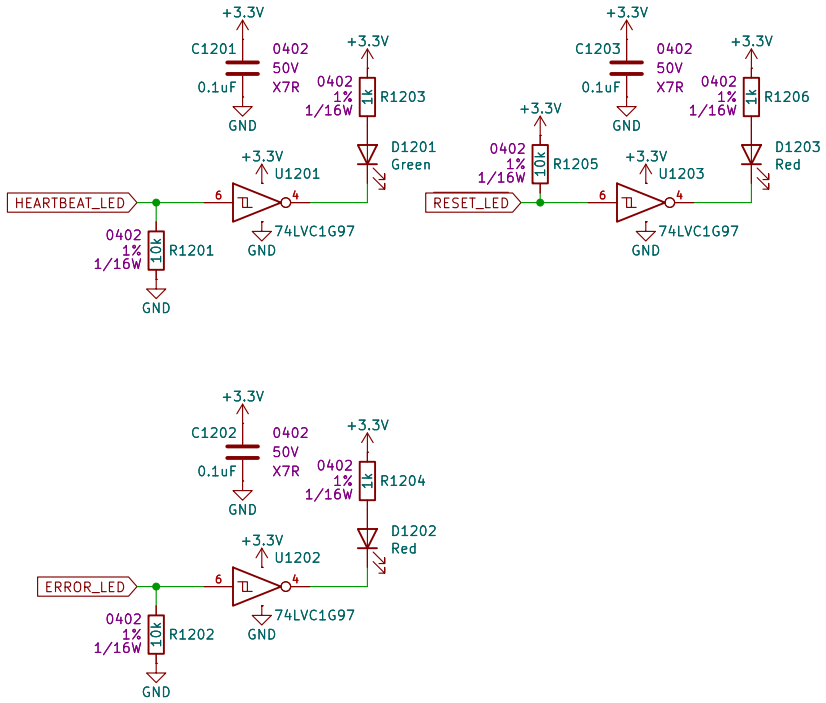
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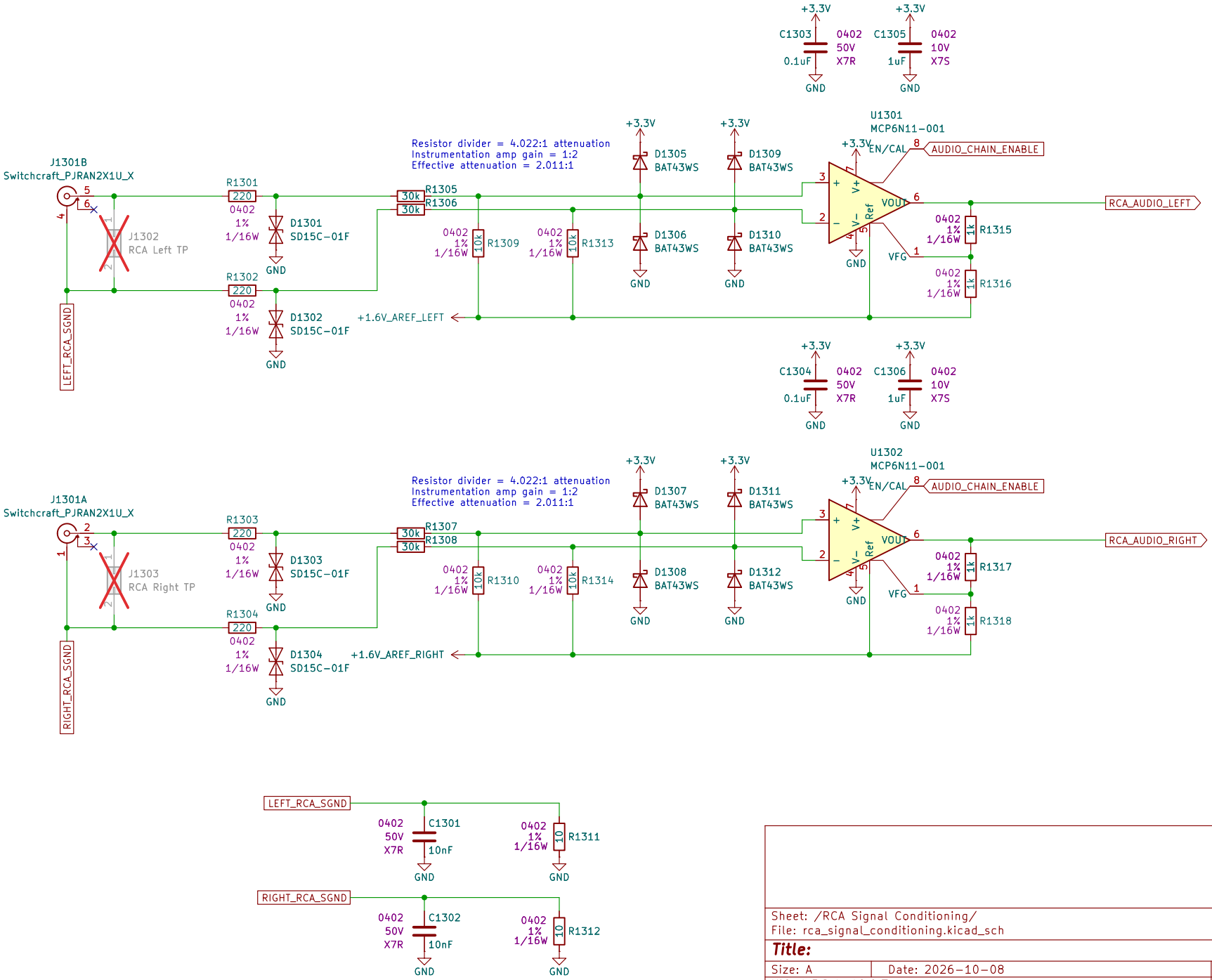
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KiCad E.D.A. 10.0.7		Id: 10/21



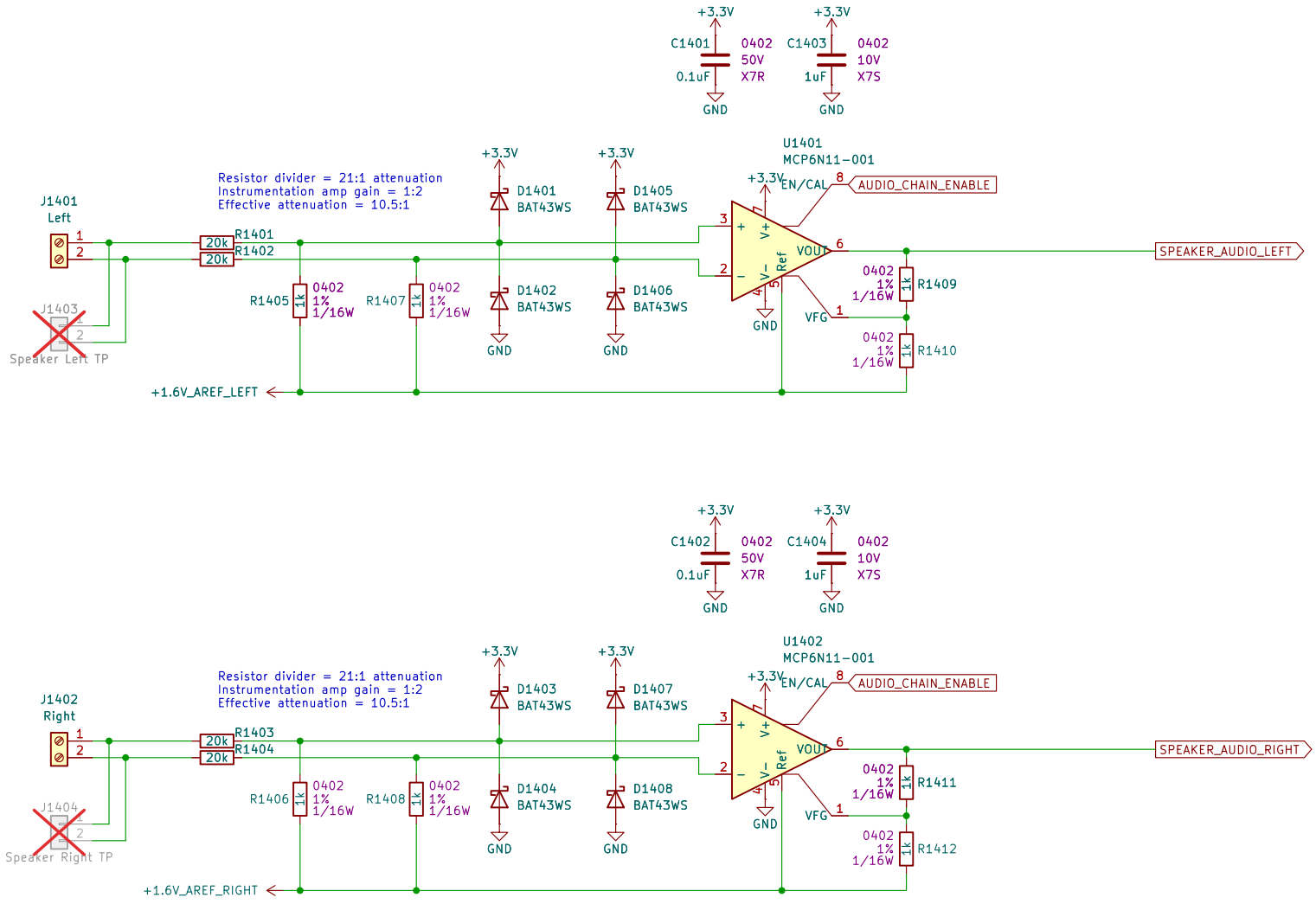
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KiCad E.D.A. 10.0.7		Id: 11/21



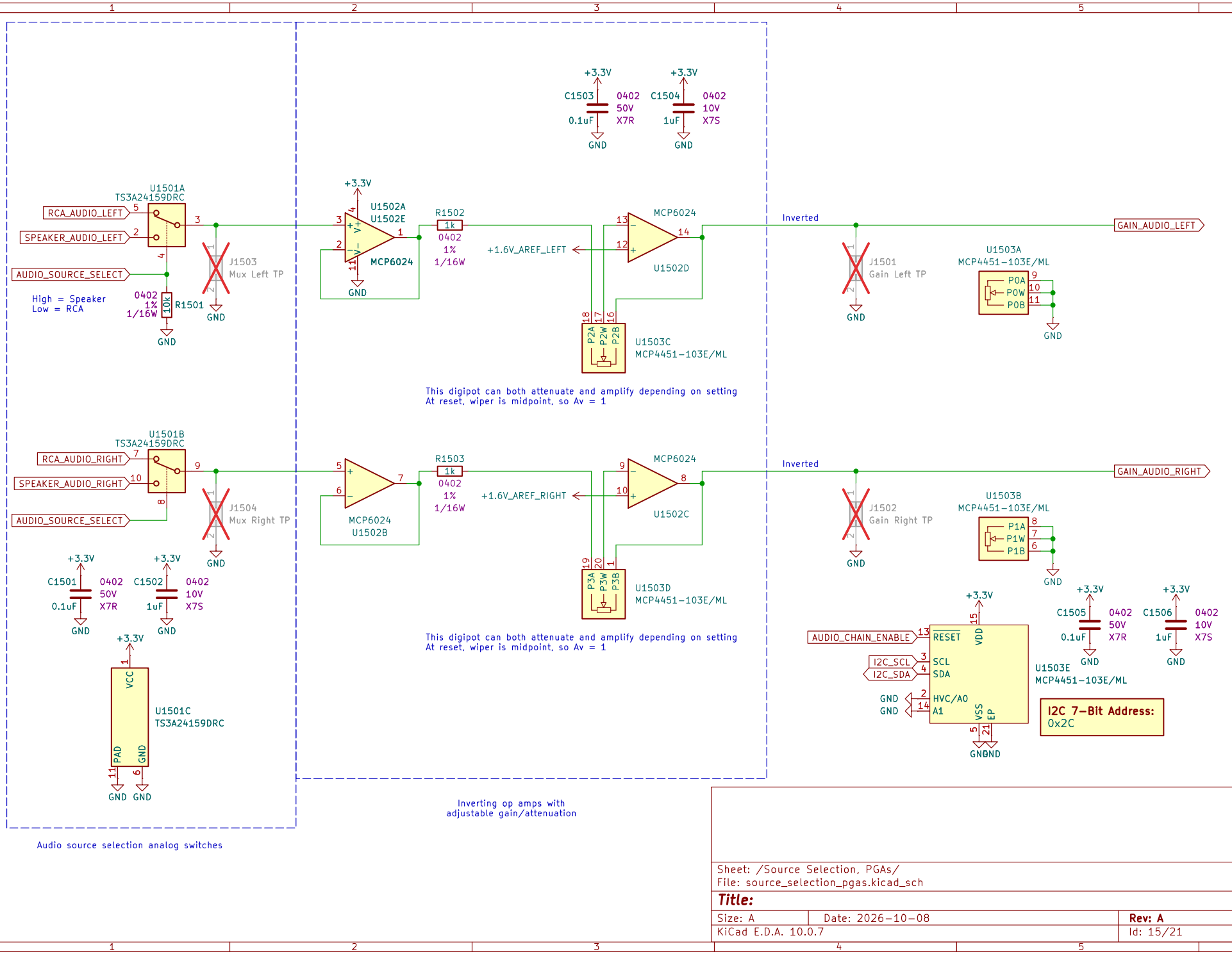
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Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 12/21



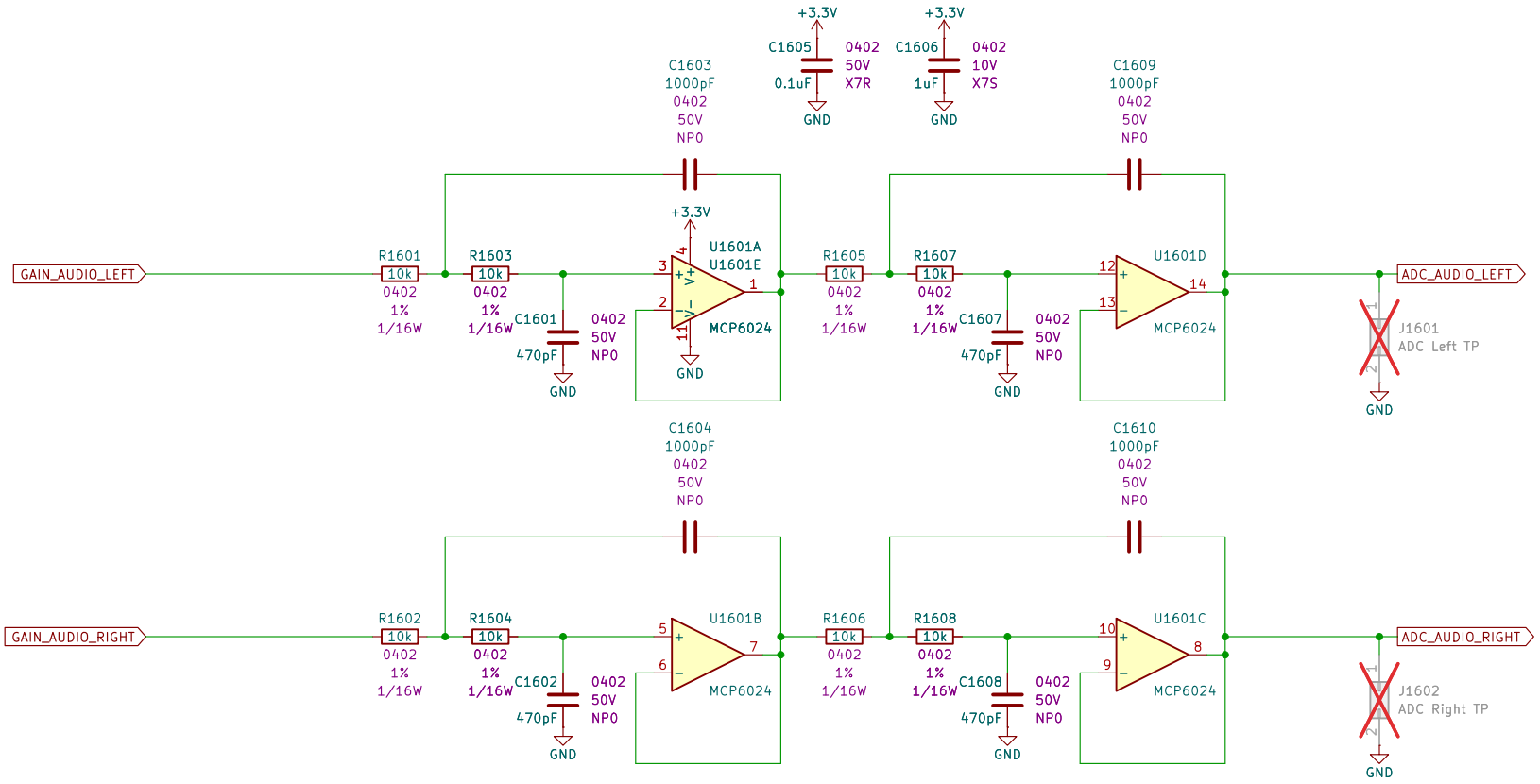
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Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 13/21



Sheet: /Speaker Signal Conditioning/		
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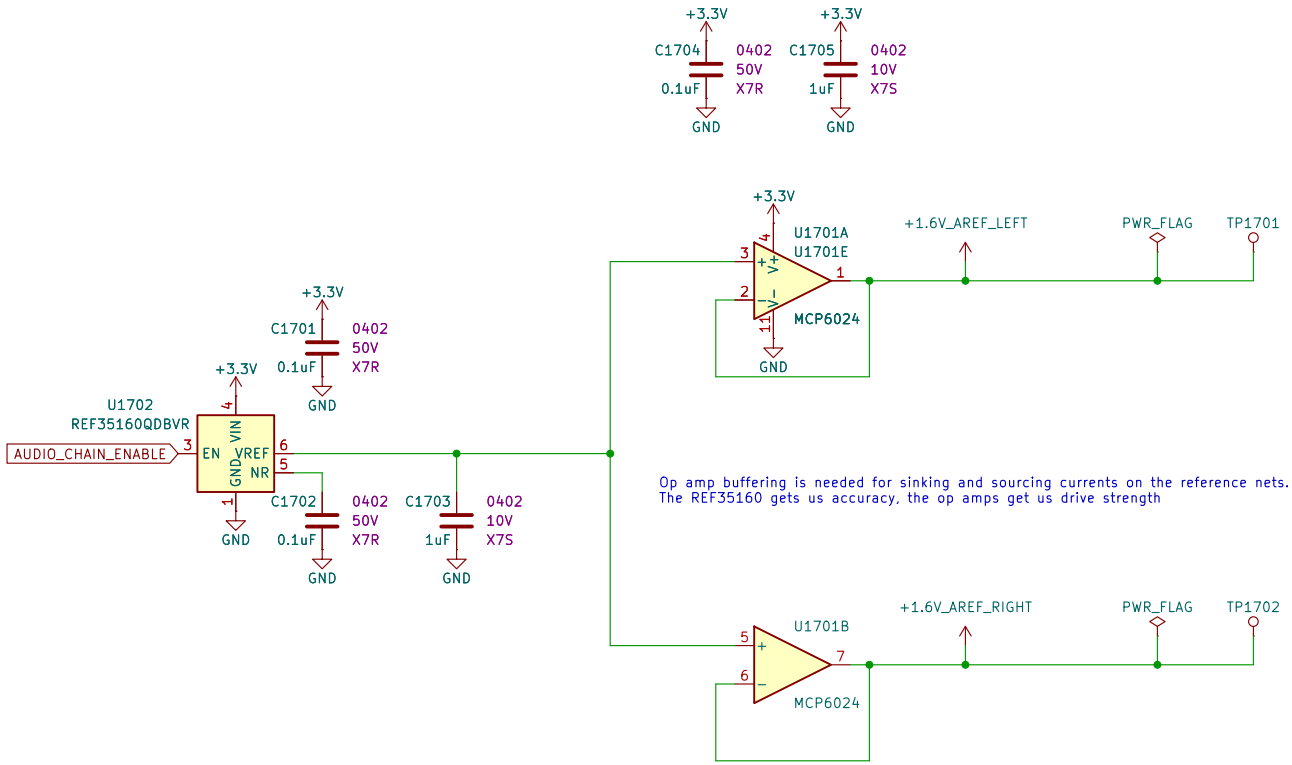


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Title:		
Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 15/21



4-pole Sallen-Key active anti-aliasing lowpass filters
Unity gain
fc = 23.2kHz, Q = 0.707 each
If sampling at 44.1kHz, nyquist is 22.05kHz
Must filter hard around this frequency

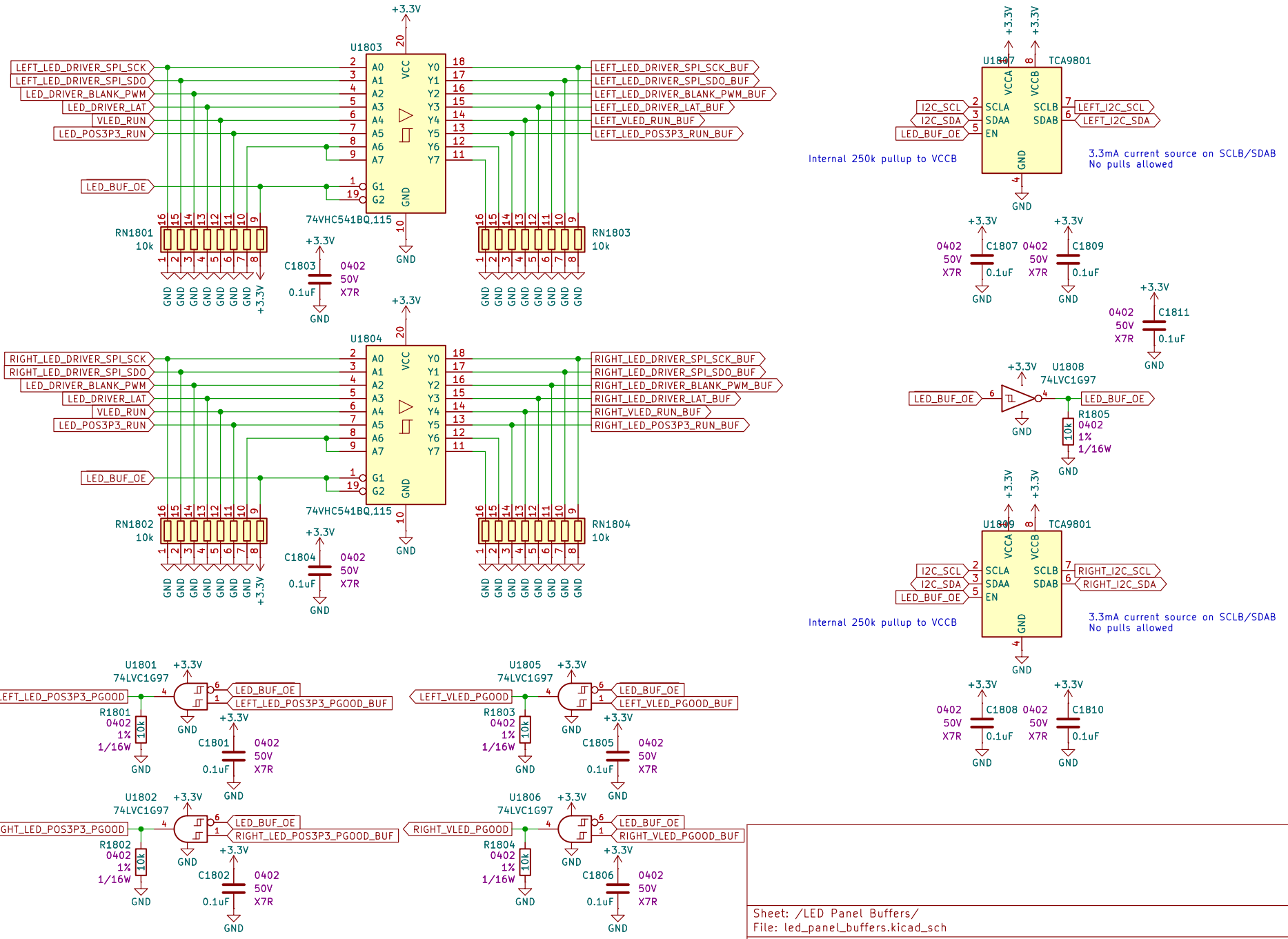
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KiCad E.D.A. 10.0.7		Id: 16/21



Sheet: /Audio Voltage References/
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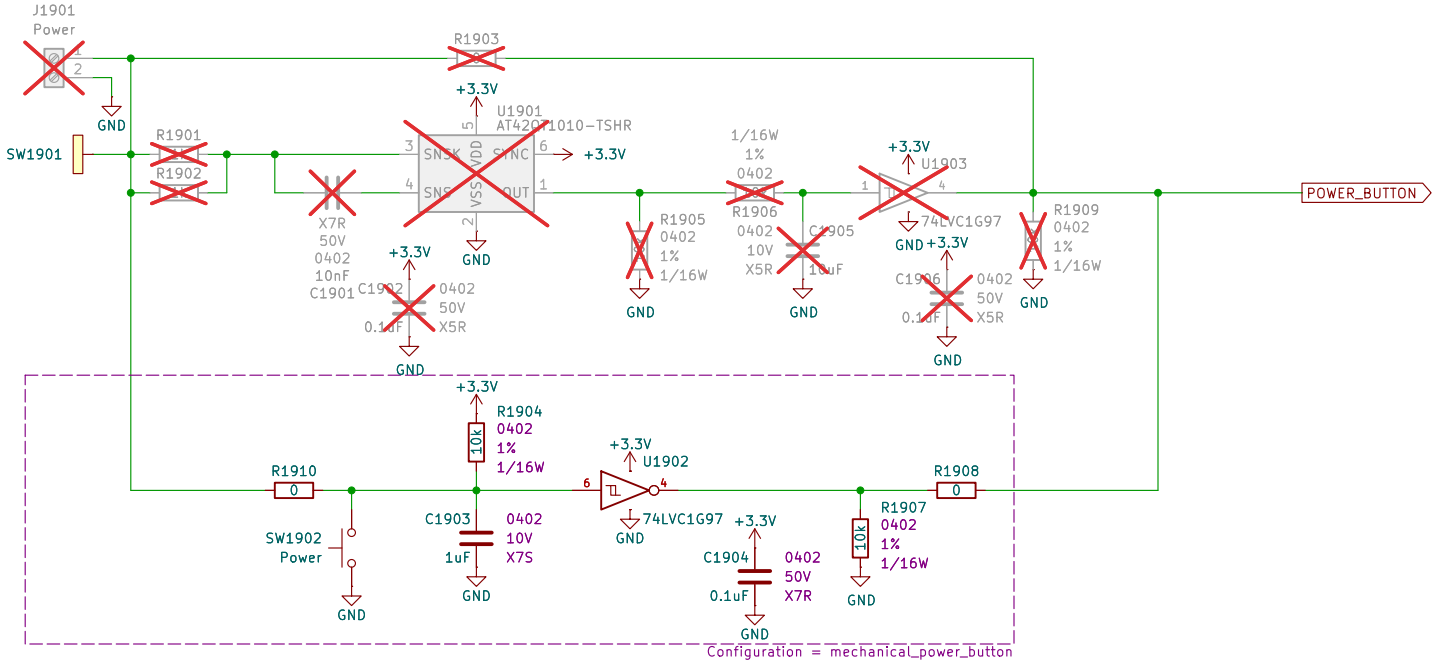
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KiCad E.D.A. 10.0.7		Id: 17/21

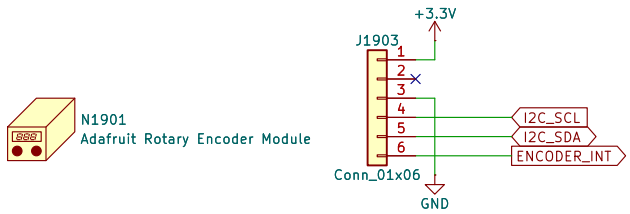


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File: led_panel_buffers.kicad_sch

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Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 18/21





I2C 7-Bit Address:
0x36

Sheet: /Pushbuttons/ File: pushbuttons.kicad_sch		
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Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 19/21

H2001
MountingHole

H2003
MountingHole

H2002
MountingHole

H2004
MountingHole

Screw
MK2001

Screw
MK2005

Screw
MK2002

Screw
MK2006

Standoff
MK2003

Standoff
MK2007

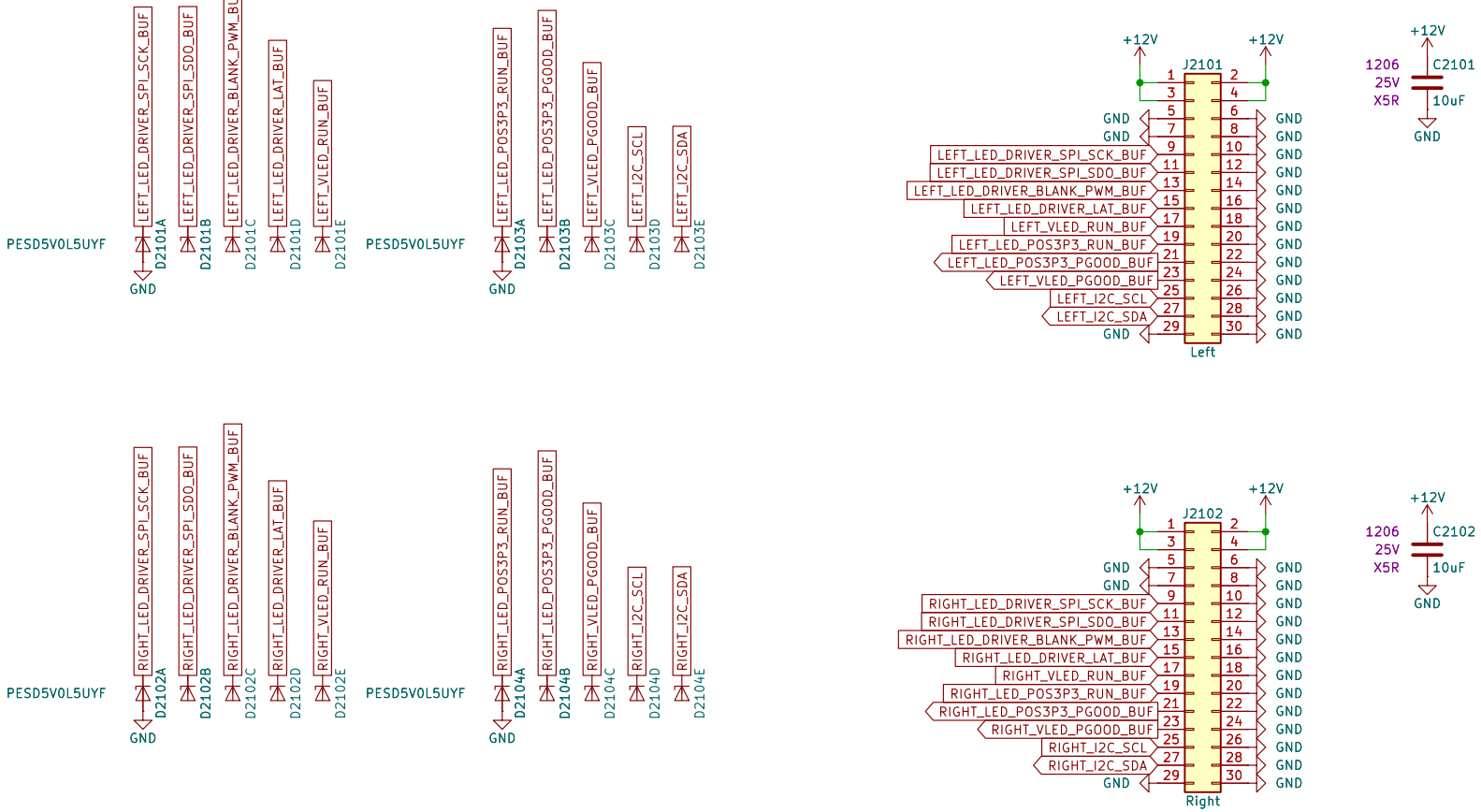
Standoff
MK2004

Standoff
MK2008

Sheet: /Mechanical/
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Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 20/21



Sheet: /LED Panel Connectors/ File: led_panel_connectors.kicad_sch		
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Size: A	Date: 2026-10-08	**Rev: A**
KiCad E.D.A. 10.0.7		Id: 21/21